2. An attorney claims that more than 25% of all lawyers advertise. A sample of 200 lawyers in a certain city showed that 63 had used some form of advertising, At $\alpha = 0.05$, is there enough evidence to support the attorney's claim? Use the p-value method.

   (a) State the hypotheses.

   $H_0: p \le 0.25$ and $H_1: p > 0.25$

   $np = 200 \cdot 0.25 = 50$

   $n(1-p) = 200 \cdot 0.75 = 150$

   (b) $\hat{p} = \frac{x}{n} = \frac{63}{200} = 0.315$

   $p = 0.25$ $\quad 1 - p = 0.75$

   $z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} = \frac{0.315 - 0.25}{\sqrt{(0.25)(0.75)/200}} = 2.12$

   (c) Find the p-value.

   $p = P(z > 2.12) = 1 - P(z < 2.12) = 1 - 0.9830 = 0.0170$

   (d) Make a decision.

   Reject Ho since $0.0170 < 0.05$ (pvalue $< \alpha$)

   (e) Make a conclusion.

   There is enough evidence to support the attorney's claim that more than 25% of the lawyers use some form of advertising.

A recent survey found that 64.7% of the population own their homes. In a random sample of 150 heads of households, 92 responded that they owned their homes. At the 0.01 level of significance, does that indicate a difference from the national proportion?

$H_0: p = 0.647$ $\quad H_a: p \neq 0.647$

$\hat{p} = \frac{92}{150} = 0.613$

$p = 0.647$ $\quad 1-p = 0.353$

$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{0.613 - 0.647}{\sqrt{\frac{0.647(1-0.647)}{150}}} = -0.87$

$\alpha = 0.01 \longrightarrow \frac{\alpha}{2} = 0.005 \Rightarrow z = \pm 2.58$

Since $|z| < z_\alpha$ we fail to reject Ho, and conclude that we cannot say that the national proportion of people owning their homes is different from 0.647.